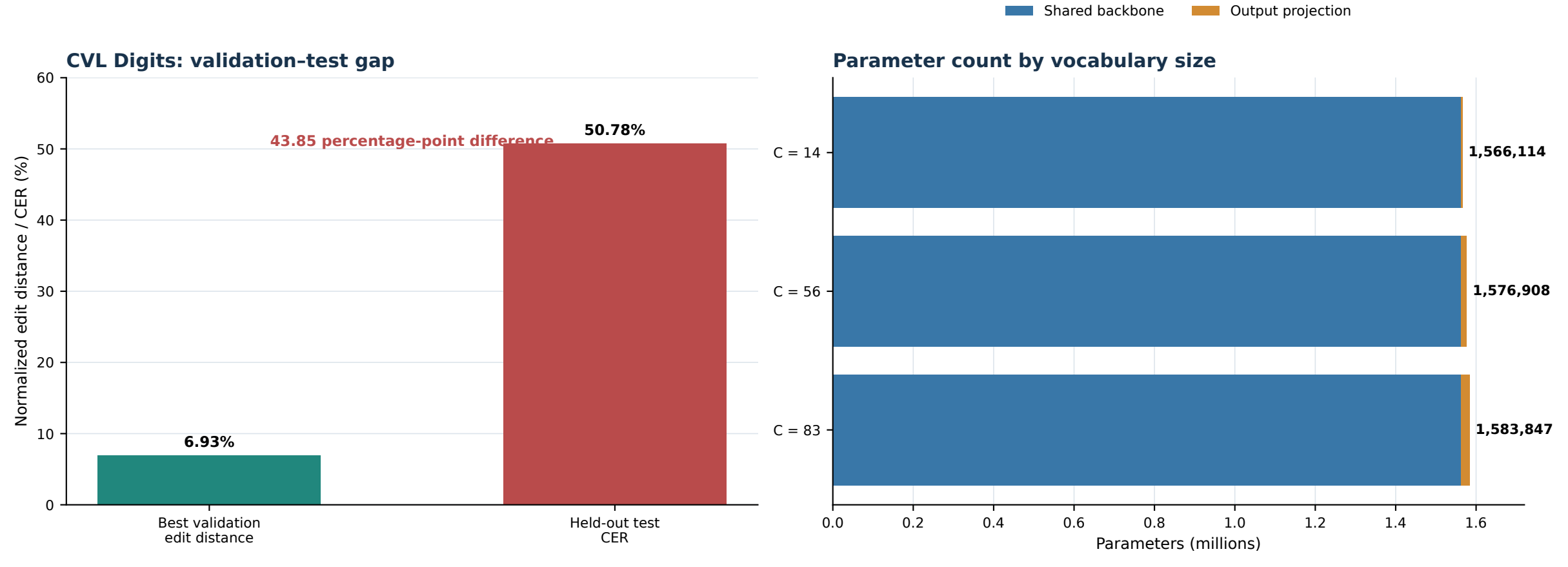DIAGNOSTIC AND MODEL-COMPLEXITY ANALYSIS